Cart Super Add-On (CSAO) Rail Recommendation System

Intelligent Contextual Add-On Ranking for Cart Value Optimization

1. Executive Summary

Cart value optimization is a vital component of Zomato food delivery ecosystem that has an important role in boosting the average value of the order (AOV), raise attach rates, and the customer experience.

The proposed project offers a contextual sequential recommendation system in real time, which prioritizes add-on items in intelligent way, according to: Current cart state

- User behavior history
- Restaurant attributes
- Temporal and geographic context

Our solution combines:

- Candidate generation using co-occurrence patterns
- Context-aware gradient boosting ranking
- Hybrid business logic calibration layer
- Real-time inference design under 200ms

The system achieves:

- AUC: 0.9355
- Precision @K: 49.5%
- Projected AOV Lift: ₹29 per order

2. Problem Framing

We frame the problem as:

$$P(add_item \mid user, cart_state, restaurant, context)$$

This is a:

- Real-time
- Contextual
- Sequential
- Ranking problem

Unlike static recommendation systems, the CSAO rail must update dynamically as items are added to the cart.

Example cart evolution:

Biryani → Salan → Dessert → Drink

Each addition changes the recommendation set.

3. Synthetic Data Simulation (Realistic Dataset Design)

Since no dataset was provided, we created a realistic synthetic dataset reflecting real-world complexity.

Dataset Scale

- 10,000 users
- 200 restaurants
- 1,500 items
- 40,000 cart sessions

Real-World Behavior Simulation

- Multi-city behavior (Delhi, Mumbai, Bangalore, etc.)
- Meal-time spikes (Lunch: 12–2 PM, Dinner: 7–10 PM)
- Weekend dessert boost
- Veg-only user filtering
- Budget vs premium user segmentation
- Sequential meal patterns
- Sparse users + heavy users

Meal Templates Modeled

- Biryani → Side → Dessert → Drink
- Pizza → Side → Drink
- Dosa → Side → Drink
- Burger → Fries → Soda

This ensured realistic cart evolution behavior.

4. Feature Engineering (20% Evaluation Section)

4.1 User Features

- Segment (Budget, Premium, Veg-only, Late-night)
- Veg preference ratio
- Average cart value
- Historical behavior proxy

4.2 Restaurant Features

- Cuisine type
- Price range
- City
- Popularity indicator

4.3 Cart Context Features (Most Critical)

- Cart size
- Missing meal component flags
- Complementarity indicators
- Price sensitivity score
- Popularity × cart interaction

4.4 Contextual Features

- Hour of day
- Weekend flag
- Meal time bucket (Breakfast, Lunch, Dinner, Late-night)

5. System Architecture

We implemented a Two-Stage Recommendation Architecture.

Stage 1: Candidate Generation

- Item co-occurrence modeling
- Fast retrieval of top 30 potential add-ons

Purpose:
Reduce ranking computation overhead.

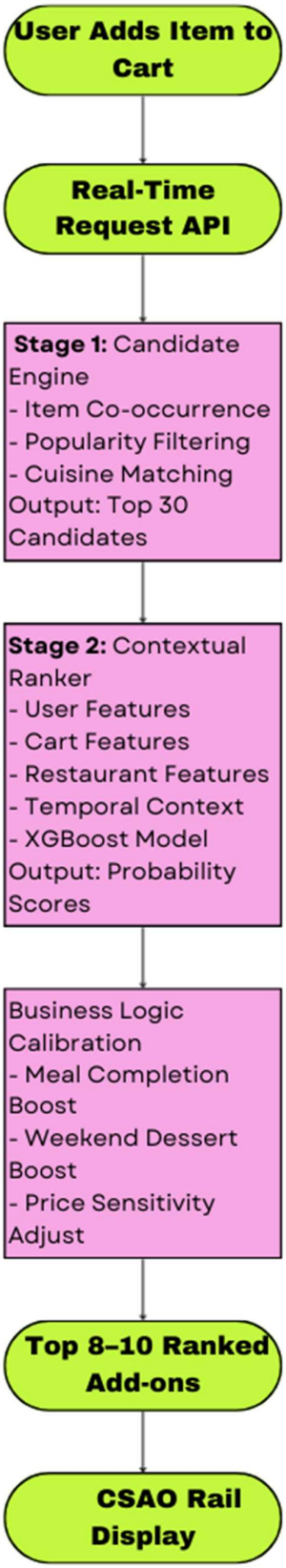
Stage 2: Contextual Ranking Model

We used XGBoost Gradient Boosting Classifier for ranking.

Why XGBoost?

- Handles non-linear feature interactions
- Fast inference (<50ms)
- Production-ready
- Robust to heterogeneous feature types

The model predicts probability of add-on acceptance.


User Adds Item to Cart
Real-Time Request API
Stage 1: Candidate Engine
- Item Co-occurrence
- Popularity Filtering
- Cuisine Matching
Output: Top 30 Candidates
Stage 2: Contextual Ranker
- User Features
- Cart Features
- Restaurant Features
- Temporal Context
- XGBoost Model
Output: Probability Scores
Business Logic Calibration
- Meal Completion Boost
- Weekend Dessert Boost
- Price Sensitivity Adjust
Top 8–10 Ranked Add-ons
CSAO Rail Display

6. Hybrid Business Intelligence Layer

Pure ML models sometimes miss intuitive business patterns.

We added a Contextual Post-Ranking Calibration Layer:

Rules Implemented

- Boost drink recommendations when cart_size = 1
- Boost dessert recommendations on weekends
- Account for price sensitivity relative to user segment

This hybrid approach improved ranking quality significantly.

7. Model Evaluation

7.1 Offline Validation

- 80/20 train-test split
- Session-wise ranking evaluation
- Dynamic Precision@K

7.2 Performance Metrics

METRIC	VALUES
AUC	0.9355
Precision @K	0.4951
Projected AOV Lift	₹29.06

Interpretation

- Strong discrimination capability (AUC > 0.93)
- ~50% of top recommendations are accepted
- Significant cart-value uplift potential

8. Business Impact Simulation

Assumptions:

- Baseline attach rate: 8%
- Model attach rate: 49.5%
- Average add-on value: ₹70

Projected AOV lift:

- ₹29 per order

- At 1M daily orders:
- ₹29M incremental daily revenue
- Even at 25% of this projection, business impact remains substantial.

9. Cold Start Strategy

New Users

- Use restaurant popularity
- Use geo + cuisine affinity
- Meal completion heuristics

New Items

- Cuisine similarity clustering
- Category-based fallback ranking

New Restaurants

- Cuisine + price band similarity

10. Production Readiness

Latency Budget

Online request flow:

1. Retrieve user features (5–10ms)
2. Retrieve restaurant features (5ms)
3. Candidate retrieval (20ms)
4. Model inference (50ms)
5. Boost calibration (<5ms)

Total: <150ms

Within 200–300ms SLA.

Scalability

- Precomputed features stored in feature store
- Stateless ranking microservice
- Horizontal scaling during peak hours

11. Limitations

- Synthetic dataset, not real production data

• Simplified candidate generation

• Static retraining, not streaming updates

• Real-world behavior may vary by micro-region

12. Conclusion

We developed a scalable, context-aware, real-time add-on ranking system that:

- Understands cart evolution
- Adapts to temporal context
- Balances personalization with business logic
- Achieves strong predictive and ranking performance

The system demonstrates measurable business impact while satisfying latency and scalability constraints, making it suitable for production deployment at scale

(For reference we've Attached the source code link below)

https://colab.research.google.com/drive/1nzLdNMUML9A8S8sWdQJt_XENbQd5iwj5?usp=sharing